HZSCM: Hyperspectral Image Zero-Shot Classification via Vision-Language Models

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***Abstract*—Most hyperspectral image (HSI) classification methods assume that all classes in the test set are present during training. However, in real-world applications, acquiring labeled training samples is challenging. As a result, it is difficult for the training dataset to cover all possible land cover types, leading to the generalized zero-shot learning (GZSL) problem. Recently, vision-language models (VLMs) have provided rich semantic priors for land cover classes, offering promising potential for GZSL. However, two fundamental gaps hinder their application to HSI classification: the task paradigm gap, arising from the difference between image-level VLMs and the pixel-level HSI classification task; and the knowledge gap, due to the inconsistency between VLM features and HSI spectral–spatial representations. To bridge both gaps, a novel framework leveraging VLM semantic priors for GZSL in HSI classification is proposed, primarily using a pseudo-labeling technique to provide knowledge for unseen classes. Specifically, a pseudo-label generation and enhancement module enables a paradigm transition from image-level understanding to pixel-level classification by incorporating HSI's spatial information. A pseudo-label correction module then refines noisy labels using spectral cues to address the knowledge gap. Finally, a global learning strategy integrates pseudo-label distillation, supervised learning, and feature regularization to classify seen classes while enabling generalization to unseen ones. Experiments on benchmark HSI datasets demonstrate the proposed method's superiority in generalized zero-shot classification. This work highlights the potential of VLMs in advancing HSI classification in practical applications.**

***Index Terms*—Deep learning, generalized zero-shot learning (GZSL), hyperspectral image (HSI) classification, vision-language model (VLM).**

I. Introduction

HYPERSPECTRAL remote sensing is a powerful imaging technology that captures fine spectral information across hundreds of wavelength channels, ranging from the visible to the near-infrared spectrum [1]. Unlike RGB or multispectral images, which capture only a limited number of broad spectral bands, a hyperspectral image (HSI) enables the precise characterization of land cover by providing detailed spectral signatures for various materials. This spectral information makes HSIs highly valuable in remote sensing applications such as environmental monitoring [2], precision agriculture [3], and urban planning [4]. The task of HSI classification involves assigning a unique class label to each pixel and is crucial for many real-world applications, ranging from land use mapping to mineral exploration [5]. HSI classification has been widely studied in the remote sensing community for many years [6], [7].

Traditional methods for HSI classification primarily focus on feature extraction using statistical learning techniques. These approaches have been extensively studied to reduce the dimensionality of hyperspectral data and to extract spectral or spectral–spatial features [8]. Classifiers like support vector machines were widely adopted to leverage these features [9]. However, the performance of these methods was limited by their reliance on hand-crafted features, which are difficult to design and often specific to particular scenarios. Moreover, they struggled to capture complex spectral–spatial relationships in HSIs, resulting in suboptimal classification performance [10].

In recent years, deep learning methods have emerged as powerful tools for HSI classification, leveraging their capacity to automatically extract high-level features [11]. Convolutional neural networks (CNNs) have been particularly successful in capturing local patterns through hierarchical feature learning [12]. Various CNN architectures, including 1-D CNNs for spectral features [13], 2-D CNNs for spatial features [14], and hybrid 3-D CNNs for joint spectral–spatial feature extraction [15], have achieved significant progress in HSI classification. Furthermore, transformer-based models have gained popularity due to their ability to model long-range dependencies, which are critical for accurate classification in complex scenes [16]. Recent advances have further combined the strengths of CNNs and transformers, integrating local convolutional operations with global attention mechanisms to effectively balance locality and long-range dependency modeling [17]. These methods have shown significant improvements over traditional machine learning approaches. However, they still operate in closed-set environments, where the set of possible classes is known a prior and fixed during training, that is, all training and test data share the same label space [18].

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However, in real-world applications, acquiring labeled training samples is challenging. And the real world is open due to environmental changes, land-use transformations, or unanticipated phenomena [19]. As a result, it is difficult for the training dataset to cover all possible land cover types. Consequently, the assumption that all land-cover types are known and available in the training dataset is often impractical [20]. During the testing phase, the model may encounter new land cover types that were not present in the training dataset, which can be described as an open-set classification (OSC) scenario [21], [22], [23]. In this case, closed-set-based classification methods tend to misclassify unknown samples into known categories. And thus, many researchers began to develop OSC methods where the classifier can accurately classify pixels from known classes and detect (or reject) pixels from unknown classes during testing [24].

Although the current OSC methods have made progress in tackling the open-set problem in HSI classification, they focus primarily on detecting pixels from unknown classes without the capability to classify them. This limitation highlights the need for methods to handle the generalized zero-shot classification problem, where the goal is not only to classify pixels from seen classes in the training set, but also to classify pixels from unseen classes that were not present during the model training process using auxiliary knowledge [25]. Unfortunately, addressing generalized zero-shot learning (GZSL) in the HSI classification remains a significant challenge. In practice, labeled HSI data are scarce and available datasets often exhibit significant variations in spectral bands, spatial resolution, spectral range, and imaging conditions due to their diverse acquisition sources. In addition, inconsistencies in class annotation standards across datasets further exacerbate distribution discrepancies, making large-scale pretraining and unified representation learning particularly difficult. These issues hinder the extraction of auxiliary information and the establishment of semantic relationships between seen and unseen classes through conventional zero-shot learning (ZSL) paradigms. Fortunately, existing visual-language models (VLMs) have demonstrated strong capabilities in learning semantics for novel categories, which can potentially aid in recognizing unseen classes without requiring manual annotations, thereby offering opportunities for zero-shot HSI classification [26]. By providing semantic priors for both seen and unseen classes in a given HSI, VLMs can serve as a bridge for knowledge transfer in zero-shot settings. However, directly applying VLMs to HSI classification remains nontrivial due to the domain gap between natural image understanding and HSI classification. This domain gap can be mainly characterized in two aspects.

1) *Task Paradigm Gap:* VLMs are typically designed for image-level understanding tasks, such as scene classification or visual question answering, based on holistic visual inputs. In contrast, HSI classification is fundamentally a pixel-level semantic segmentation task, requiring dense labeling of each pixel in the given HSI. This discrepancy in task formulation and input–output granularity leads to a substantial mismatch, making it challenging to directly utilize VLMs for pixel-wise classification in HSI. Bridging this task paradigm gap is difficult but essential to enable VLMs to provide semantic priors for unseen classes at the pixel level.
2) *Knowledge Gap:* VLMs are pretrained on large-scale RGB image–text pairs and excel at capturing high-level semantic representations, such as object categories and contextual relationships. However, HSI classification relies on fine-grained spectral signatures and spatial-contextual patterns that are domain-specific and not well represented in natural image datasets. A key challenge, therefore, lies in effectively adapting and aligning the general semantic knowledge learned by VLMs with the spectral and spatial characteristics unique to HSI data.

To address the aforementioned challenges, we propose a novel HSI zero-shot classification method, named HZSCM, that bridges the task paradigm gap and knowledge gap between VLMs and HSI. The proposed approach integrates semantic priors from VLMs with spectral–spatial characteristics of HSI to achieve GZSL for HSI classification through a three-stage pipeline: First, to tackle the task paradigm gap, we develop a superpixel-based pseudo-label generation strategy that enables a paradigm transition from image-level VLM understanding to pixel-level HSI classification. By combining superpixel segmentation, text templates, and semantic matching from VLMs, we generate initial pixel-level pseudo-labels for both seen and unseen categories. To enhance spatial consistency and accuracy, we further incorporate the segment anything model (SAM) [27] and ensemble learning. Second, to mitigate the knowledge gap, we introduce a spectral-guided label correction mechanism that leverages the intrinsic spectral coherence of HSI data. By incorporating spectral consistency constraints, this method ensures the semantic knowledge transferred from VLMs is effectively aligned with the spectral–spatial characteristics of HSI. Finally, the corrected pseudo-labels, along with available supervised annotations, are utilized in a global learning stage using a standard semantic segmentation network (e.g., U-Net). This integration enables effective classification across both seen and unseen categories.

The main contributions are summarized as follows.

1) This article presents a novel HSI zero-shot classification framework that adapts VLMs to provide pixel-level semantic priors, bridging the task paradigm gap (VLM image-level understanding vs. HSI pixel-level labeling) and knowledge gap (VLM semantics vs. HSI spectral–spatial features).
2) We design a superpixel-based pseudo-label generation method for VLMs. It innovatively generates pixel-level pseudo-labels for HSI land-covers, capturing unseen-category semantics. It also integrates SAM and ensemble learning to enhance pseudo-label spatial coherence and accuracy.
3) A label correction mechanism based on spectral information is proposed. It aligns pseudo-labels with HSI's unique spectral–spatial characteristics, refining VLM-transferred semantic knowledge and ensuring pseudo-label accuracy and consistency.

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The remainder of this article is organized as follows. Section I reviews related works. Section III introduces the proposed zero-shot classification framework and details its components. Section IV evaluates the performance of the proposed HZSCM in comparison with state-of-the-art approaches and presents the experimental results. Finally, Section V concludes the article and outlines directions for future research.

II. Related Works

A. Deep Learning Methods for HSI Classification

Deep learning models, particularly CNNs and transformers, have driven substantial progress in HSI classification by leveraging their strong ability in learning spectral and spatial representations. Early CNN-based methods primarily extracted spectral features using 1-D convolutions [28], and were soon extended to 2-D and 3-D architectures to exploit spectral–spatial correlations [29], [30]. To further enhance representation capacity, various strategies were proposed, including the incorporation of geometric algebra [31], spatial constraint losses [32], and attention mechanisms [33]. Hybrid CNN frameworks that integrate graph convolutional networks (GCNs) have also shown effectiveness in fusing local and global features [34]. In addition, CNNs have been enhanced with transfer learning, domain adaptation, and few-shot learning to address challenges such as data scarcity and domain shift [35], [36], [37].

Recently, transformer-based methods have emerged as powerful alternatives for HSI classification due to their self-attention mechanism, which excels at modeling long-range dependencies in spatial-spectral features [38]. For example, HSI-BERT [39] treated each pixel as a token to capture global contextual information. Recent models further explore cross-domain alignment and sample efficiency via encoder–decoder architectures, cross-attention mechanisms, and knowledge distillation [40], [41], [42]. Moreover, hybrid networks that combine CNNs and transformers have been proposed to integrate local detail extraction with global context modeling, achieving state-of-the-art performance on multiple benchmarks [43], [44].

Despite their success, the above methods are developed under the assumption that the training and testing sets share the same label space. However, scenarios where this assumption does not hold remain largely unexplored, highlighting the need for research on HSI zero-shot classification.

B. Open Set Learning for HSI Classification

Recent advancements in open-set learning for HSI classification have made significant progress in addressing the limitations of closed-set assumptions by introducing strategies to handle unknown classes effectively. Prototype-based methods, such as the feature-invariance network with weighted Pearson distance measurement [45] and the feature consistency network with contrastive clustering module [46], utilized prototype-guided modules, and novel distance metrics to enhance class separation and identify unknown samples. Reconstruction-based approaches, such as spectral–spatial latent reconstruction frameworks [47], improve robustness by simultaneously reconstructing spectral and spatial features to retain useful information for rejecting unknown classes while preserving accuracy for known classes. Advanced architectures, including fractional Fourier-based networks [48] and MLP-like models with reciprocal points learning [49], further enhance open-set recognition by expanding feature representations, modeling extra-category spaces, and introducing dynamic thresholds to distinguish unknown categories. Moreover, meta-learning techniques like extreme value meta-learning leverage Weibull distributions and open quadruplet loss to maintain topological consistency, enabling fine-grained detection of unknowns and improving generalization across different datasets [50]. However, these methods remain limited to identifying unknown classes without providing the classification capabilities or generalization to unseen categories that ZSL approaches achieve.

C. ZSL in Remote Sensing

ZSL aims to classify unseen classes during testing by transferring knowledge from seen classes in the training set. GZSL extends this to classify both seen and unseen classes in test scenarios, ensuring balanced performance across all categories [51]. The classes' semantic embeddings (or auxiliary information) that connect seen and unseen categories are mainly used to achieve ZSL. These embeddings, which can be derived from predefined attributes [52], or word embeddings [53] learned from sources like Wikipedia or large amounts of documents, enabling connections between visual features and semantic information, thereby facilitating knowledge transfer [54]. ZSL typically maps visual features and semantic embeddings into a shared latent space, allowing samples to cluster around their corresponding semantic descriptors. During inference, labels for unseen classes are assigned by identifying the nearest semantic descriptor in this space [55]. Research has focused on refining embedding spaces [56] and improving compatibility functions [57]. Generative approaches, such as GANs and VAEs, have also been explored to synthesize features for unseen classes, reframing ZSL as a standard classification task [58].

In remote sensing, ZSL has been primarily applied to scene classification tasks [59], [60]. Early studies employed word embeddings like Word2Vec to derive semantic features and construct relational graphs between categories [61]. Subsequent work introduced semi-supervised alignment strategies to improve knowledge transfer [62]. Recent advances include distance-constrained semantic autoencoders [63], transformer-based methods leveraging pretrained language models such as BERT [64], and GAN-based approaches for generating visual features from semantic representations [65], all aimed at bridging the domain gap between seen and unseen categories. More recently, VLMs have emerged as a promising direction for ZSL in remote sensing [66]. Xu et al. [67] designed an automatic attribute annotation scheme using CLIP and utilized transformers to associate local image regions while incorporating contextual background for more accurate predictions. Despite these promising developments in remote sensing scene classification, zero-shot methods tailored for HSI classification

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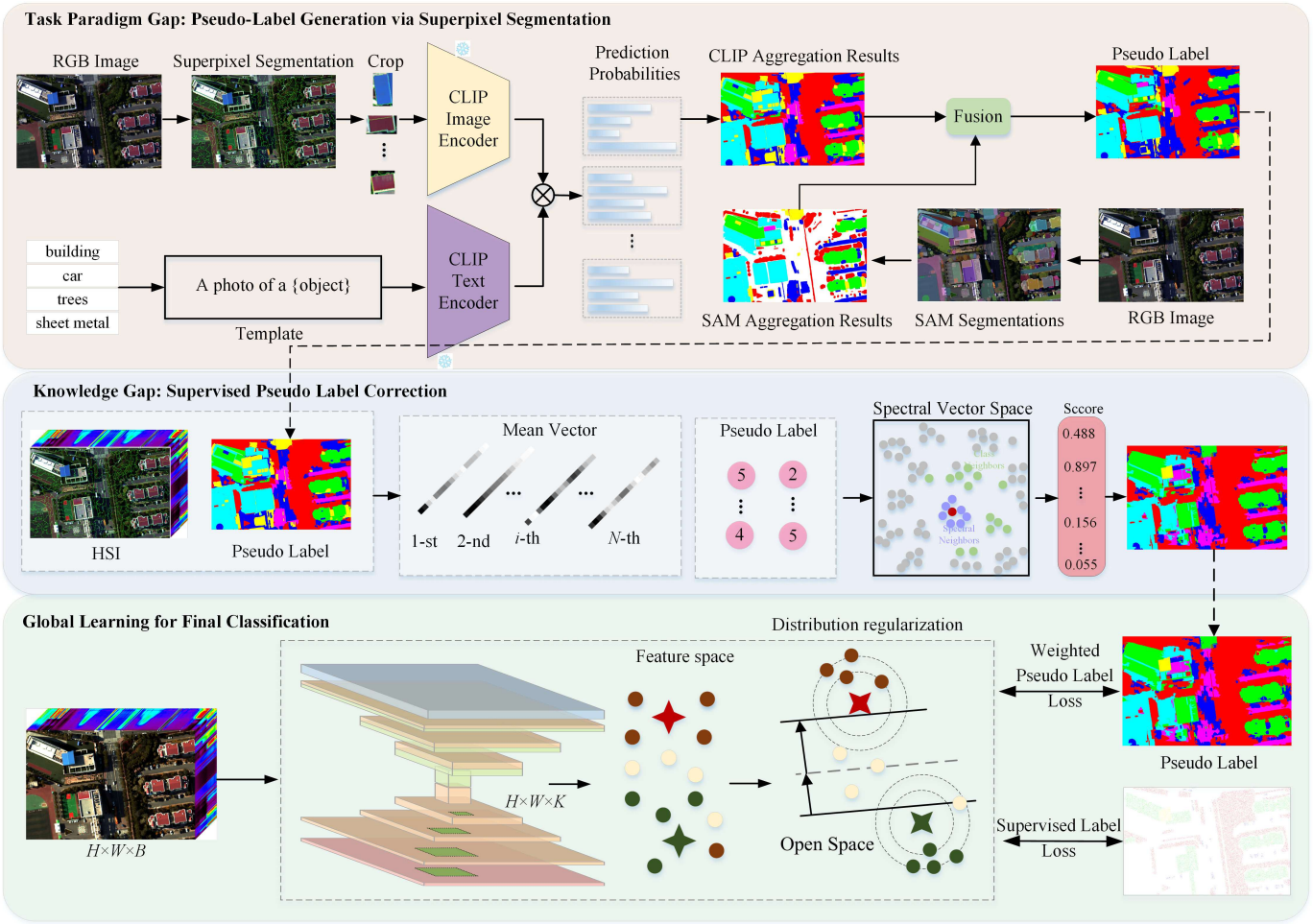


Fig. 1. Proposed ZSL framework for HSI classification overview—training flow.

remain underexplored, highlighting an urgent need for dedicated research in this area.

D. VLM in Remote Sensing

In recent years, VLMs have shown remarkable success in natural image understanding by leveraging large-scale image–text paired datasets. Motivated by this progress, the remote sensing community has begun to explore VLMs to enhance the semantic interpretation of remote sensing imagery [68]. Early studies focused on specific tasks such as image captioning [69], [70], change captioning [71], visual question answering [72], [73], visual grounding [74], [75], and text-to-image retrieval [76], [77], often using limited remote sensing image–text data. However, due to the scarcity of large-scale curated image–text pairs—since textual annotations cannot be crowd-sourced as in natural images—most existing models remain task-specific and data-constrained.

Recent efforts have aimed to overcome these limitations by constructing large-scale remote sensing vision-language datasets and foundation models. For instance, SkyScript automatically links remote sensing images with OpenStreetMap tags to create a semantically rich dataset, enabling robust zero-shot classification and retrieval [78]. RS5M similarly expands the dataset scale by filtering existing corpora and augmenting them with auto-generated captions, facilitating the fine-tuning of models such as GeoRSCLIP for improved zero-shot and cross-modal performance [79]. RemoteCLIP further enhances scalability through unified image-caption formatting and multisource data integration, supporting diverse downstream tasks like image classification, retrieval, and object counting [80]. In parallel, methods such as RSGPT [81], GeoChat [82], and SkyEyeGPT [83] introduce large multimodal language models tailored for remote sensing tasks through instruction tuning and visual-language alignment. These models demonstrate competitive performance across various tasks like zero-shot classification, grounding, and retrieval. In HSI classification, [84] improved domain generalization-based HSI classification through visual-linguistic representation alignment.

III. Methodology

As illustrated in Fig. 1, the proposed framework of HZSCM consists of three stages: unsupervised pseudo-label generation, spectral-guided label correction, and global training with knowledge transfer. First, VLMs generate initial pseudo-labels to bridge the gap between image-level understanding and pixel-wise classification. Then, these labels are refined through spectral-guided correction to enhance reliability,

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thereby addressing the knowledge gap. Finally, the corrected pseudo-labels, along with supervised labels, are used to guide global model training and facilitate effective knowledge transfer.

Specifically, object-aware superpixel segmentation is first employed to enable the paradigm transition from image-level VLM inference to pixel-level HSI classification, generating initial pseudo-labels for both seen and unseen categories. To enhance spatial coherence, the SAM and ensemble inference are further integrated. Second, in the label correction stage, a spectral-guided refinement mechanism is introduced to suppress unreliable labels by leveraging the intrinsic spectral consistency of HSI data. Finally, in the global training stage, the segmentation network is trained with the corrected pseudo-labels along with supervised training labels, where knowledge distillation and distribution regularization are applied to facilitate knowledge transfer and improve generalization to unseen classes.

A. Preliminaries

We first formalize the problem of ZSL in HSI classification. Given an HSI $\mathbf{X}$, all labeled pixels and their corresponding labels form a dataset D, which is partitioned into a training set D_{tr} for model training and a test set D_{te} for evaluation. The training set contains Y_{tr} classes and the test set contains Y_{te} classes. In addition, the known or seen classes mean the classes that are offered to model for training, denoted by Y_S, and we have $Y_{\text{tr}} = Y_S$. The classes in the test set but absent from the training set are referred as unknown or unseen classes, denoted by Y_U. In ZSL, $Y_{\text{te}} = Y_U$, indicating that the test set is composed of samples from unknown classes. In generalized ZSL, $Y_{\text{te}} = Y_S \cup Y_U$, meaning the test set comprises samples from both known and unknown classes. GZSL is more challenging due to the co-occurrence of seen and unseen categories in complex scenes. In the remainder of this article, we adopt the generalized zero-shot setting as the default.

To support category-level semantic reasoning, we leverage the contrastive language–image pretraining (CLIP) model, a representative VLM pretrained on large-scale image–text pairs [85]. Notably, alternative VLMs such as GeoRSCLIP or Florence can be seamlessly integrated into the proposed framework. CLIP comprises an image encoder f^I and a text encoder f^T, which project image and text inputs into a shared embedding space. During training, CLIP aims to predict the correct alignment between image–text pairs. Given a batch of N_b image–text pairs, the model computes a similarity matrix $S \in \mathbb{R}^{N_b \times N_b}$, where each row corresponds to the similarity between an image and all N_b text descriptions. CLIP is trained to maximize the similarity for positive pairs (correct image–text match) and minimize it for negative pairs (incorrect matches), using a symmetric cross-entropy loss function.

For a downstream classification task, given a test image $\mathbf{x}$ and K candidate classes, the model first uses the provided template to organize each class into the corresponding text $\{y_i^T\}_{i=1}^K$, such as "a photo of [CLASS]." The model then computes the corresponding image and text vectors

$$\mathbf{f}^I = f^I(x) \tag{1}$$

$$\mathbf{f}_i^T = f^T\left(y_i^T\right), \quad i = 1, \ldots, K. \tag{2}$$

Next, the cosine similarity between the text and image vectors is computed, followed by softmax normalization to obtain the prediction probability for each class

$$p_i = \frac{\exp\left(\langle \mathbf{f}^I, \mathbf{f}_i^T \rangle / \tau\right)}{\sum_{j=1}^K \exp\left(\langle \mathbf{f}^I, \mathbf{f}_j^T \rangle / \tau\right)} \tag{3}$$

where $\langle \cdot, \cdot \rangle$ denotes the calculation of cosine similarity, and τ is the temperature parameter learned during training. The predicted label corresponds to the class with the highest probability. For convenience, the inference process can be expressed as

$$\mathbf{p} \in \mathbb{R}^K = \text{CLIP}\left(\mathbf{x}, \left\{y_i^T\right\}_{i=1}^K\right) \tag{4}$$

where $\mathbf{p}$ represents the predicted probability vector.

Due to its ability to infer novel categories via textual semantics, CLIP is well-suited for ZSL tasks. Moreover, CLIP models trained on remote sensing image–text datasets, such as the GeoRSCLIP model and the RemoteCLIP model, have also shown promising performance in remote sensing tasks.

B. Unsupervised Pseudo-Label Generation

To address the absence of labeled samples for unseen classes during training, we leverage the CLIP model to generate pseudo-labels for HSI, thereby injecting prior semantic knowledge into the classification process. Considering the paradigm gap between CLIP's image-level understanding and the pixel-level nature of HSI classification, we propose a superpixel-based pseudo-label generation strategy to bridge this gap and facilitate the paradigm transition. To further improve the spatial consistency and accuracy of the pseudo-labels, we integrate the SAM and ensemble learning with multiple CLIP models.

Pseudo-Label Generation via Superpixel Segmentation. Since the CLIP model requires RGB image inputs, we first divided the HSI $\mathbf{X} \in \mathbb{R}^{H \times W \times B}$ into several superpixel segments using a superpixel segmentation technique

$$(s_i)_{i=1}^N = \text{Seg}^{\text{ERS}}(\mathbf{X}) \tag{5}$$

where s_i means the ith superpixel, N is the number of superpixel segments, and ERS refers to the entropy rate superpixel (ERS) algorithm used for segmentation [86]. Each segment contains the row and column indices of the pixels it covers. To construct a CLIP-compatible input, we first convert the original HSI into a RGB visualization $\mathbf{X}^{\text{RGB}} \in \mathbb{R}^{H \times W \times 3}$. The RGB version is generated by selecting three spectral bands corresponding to the wavelengths of red, green, and blue. For the ith superpixel s_i, we crop a rectangular patch $\mathbf{a}_i$ from the RGB visualization of HSI, where the window fully covers s_i. To avoid tight crops that lose spatial context, we enlarge the cropping window by a margin ratio α. This process can be formulated as

$$\mathbf{a}_i = \text{crop}\left(\mathbf{X}^{\text{RGB}}, s_i, \alpha\right). \tag{6}$$

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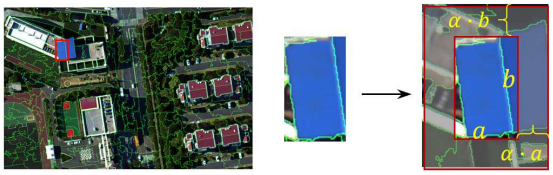
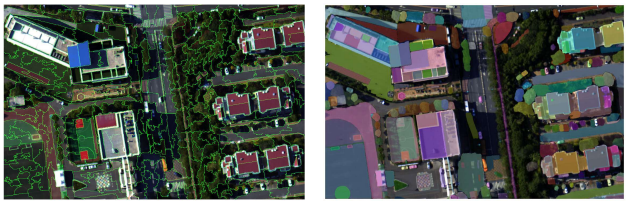


Fig. 2. Brief illustration of cropping with margin.



(a) (b)

Fig. 3. Segmentation results on the QUH-Qingyun dataset: (a) ERS and (b) SAM.

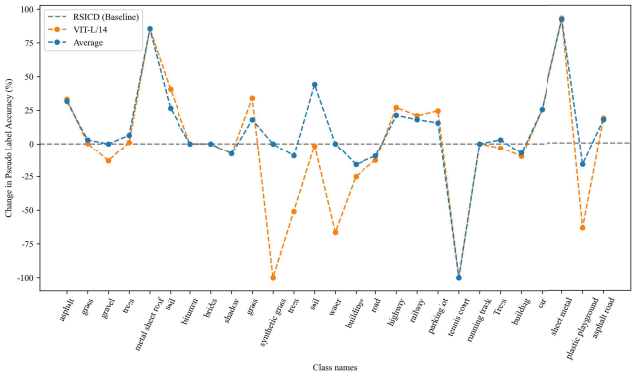


Fig. 4. CLIP models utilizing different vision encoders exhibit distinct class preferences. This phenomenon is analyzed using three widely used HSI datasets, which will be described in detail in Section IV. Specifically, we evaluate two CLIP models (i.e., RSICD-CLIP and ViT-L-CLIP) by calculating the pseudo-label accuracy for each class. Besides, their average accuracy (AA) for each class is also illustrated.

As illustrated in Fig. 2, this margin ensures sufficient background context for semantic recognition.

Next, we construct class-level text prompts $\{y_k^T\}_{k=1}^K$, where $K = Y_S \cup Y_U$, and then feed each image patch $\mathbf{a}_i$ and the prompts into the CLIP model for prediction

$$\mathbf{p}_i^{\mathrm{ERS}} = \mathrm{CLIP}\left(\mathbf{a}_i, \{y_k^T\}_{k=1}^K\right). \tag{7}$$

This process is illustrated in Fig. 1.

The resulting prediction vector $\mathbf{p}_i^{\mathrm{ERS}}$ is assigned to all pixels within the superpixel s_i, yielding a pseudo-label map for the entire HSI, denoted as $\mathbf{P}^{\mathrm{ERS}} \in \mathbb{R}^{H\times W\times K}$

$$\mathbf{P}^{\mathrm{ERS}}(h,w) = \mathbf{p}_i^{\mathrm{ERS}}, \quad \text{for all } (h,w) \in s_i. \tag{8}$$

Pseudo-Label Enhancement Based on SAM Segmentation. As shown in Fig. 3, the ERS segmentation tends to produce irregular and overly fine regions, which fails to preserve the overall semantic structure of the land-cover objects. This increases the difficulty for CLIP to recognize them and leads to low accuracy in the generated pseudo-labels. To enhance the superpixel segmentation results, we employ the SAM model for correction. The SAM is a deep learning-based image segmentation model that provides improved object-level segmentation by predicting masks from a grid of spatial prompts.

Following the same CLIP-based labeling process as above, we obtain another SAM-enhanced pseudo-label map $\mathbf{P}^{\mathrm{SAM}} \in \mathbb{R}^{H\times W\times K}$. However, SAM may not cover all pixels, and low-confidence segments are discarded. Consequently, SAM provides better segmentation results, but some pixels remain unlabeled, and their pseudo-labels are set to zero vectors. Therefore, we fill missing labels in $\mathbf{P}^{\mathrm{SAM}}$ using the corresponding ERS pseudo-labels as a complement. This process is expressed as follows:

$$\mathbf{P}^{\mathrm{ERS}}[\mathbf{P}^{\mathrm{SAM}} \neq 0] = 0 \tag{9}$$

$$\mathbf{P}^{\mathrm{CLIP}} = \mathbf{P}^{\mathrm{ERS}} + \mathbf{P}^{\mathrm{SAM}} \tag{10}$$

where $\mathbf{P}^{\mathrm{CLIP}}$ represents the integrated pseudo-labels from both segmentation results using the same CLIP. This fusion strategy balances the complete coverage of ERS with the semantic integrity of SAM, resulting in a more accurate and spatially coherent pseudo-label map $\mathbf{P}^{\mathrm{CLIP}}$.

Pseudo-Label Enhancement Based on Dual CLIP Integration. CLIP models with different encoders may exhibit diverse recognition pReferences. As shown in Fig. 4, different CLIP variants perform inconsistently across categories. To utilize the strengths of different CLIP models and thus further refine pseudo-labels, we adopt an ensemble strategy using two CLIP models, i.e., natural ViT-L-CLIP, based on OpenAI's ViT-Large backbone trained on the natural RGB images-text dataset, and RSICD-CLIP, a version fine-tuned on the RSICD remote sensing dataset. After generating pseudo-labels independently using the two models following the aforementioned process, we obtain the pseudo-labels $\mathbf{P}^{\mathrm{NAT}}$ and $\mathbf{P}^{\mathrm{RSICD}}$, respectively. The final pseudo-labels are generated by simply averaging the two models' predictions

$$\mathbf{P} = \left(\mathbf{P}^{\mathrm{NAT}} + \mathbf{P}^{\mathrm{RSICD}}\right)/2. \tag{11}$$

This dual-model integration improves robustness and reduces bias from individual CLIP models, further enhancing the quality of pseudo-labels used in the subsequent stages.

C. Supervised Pseudo-Label Correction

While pseudo-labels generated by VLMs offer a semantic initialization, the absence of hyperspectral domain knowledge leads to inevitable label noise. To mitigate the knowledge gap and reduce label noise, we propose a supervised correction method that further refines pseudo-labels using spectral similarity and limited ground-truth supervision.

To facilitate the correction process and reduce the number of pixels to be processed individually, we perform superpixel segmentation on the HSI. This step is based on the assumption that pixels within the same superpixel share the same class label, based on the fact that superpixels are homogeneous regions. Let $(s_i)_{i=1}^N$ represent the superpixel index set. For each

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superpixel s_i, its mean spectral feature $\mathbf{v}_i$ and its mean pseudo-label probabilities $\mathbf{p}_i$ is computed as

$$\mathbf{v}_i = \frac{1}{|s_i|} \sum_{(h,w)\in s_i} \mathbf{X}(h,w), \mathbf{p}_i = \frac{1}{|s_i|} \sum_{(h,w)\in s_i} \mathbf{P}(h,w). \tag{12}$$

Next, we classify superpixels into labeled and unlabeled categories. A labeled superpixel contains at least one training pixel with a label, while an unlabeled superpixel consists entirely of unlabeled pixels. For labeled superpixels, the supervised label l_i is determined by majority voting

$$l_i = \arg\max_k \left(\sum_{(h,w)\in s_i} \delta(g_{\mathrm{hw}}, k) \right) \tag{13}$$

where g_{hw} denotes the training label at pixel (h,w), $\delta(\cdot)$ is the Kronecker delta function, and $|s_i|$ is the number of pixels in superpixel s_i. For unlabeled superpixels, we assign a label of -1 to them. Since the labeled training set is very small, most of the superpixels are unlabeled.

To refine the pseudo-labels, we initially assign each superpixel a preliminary pseudo-label

$$p_i = \arg\max_k (\mathbf{p}_i). \tag{14}$$

For the ith unlabeled superpixel, we further define two types of neighborhood relationships: spectral neighbors and class neighbors. Spectral neighbors $\mathcal{S}_i$ comprise superpixels with similar spectral features, typically those grouped within the same cluster in the spectral feature space. The class neighbors $\mathcal{C}_i$ represent the superpixels which share the same pseudo-label with the given superpixel. These neighborhood relationships help identify and correct erroneous pseudo-labels.

Following [87], we define the probability that a given feature $\mathbf{v}$ is recognized as the nth sample in a given set as

$$P(n|\mathbf{v}) = \frac{e^{sim(\mathbf{v}n,\mathbf{v})}}{\sum m e^{sim(\mathbf{v}_m,\mathbf{v})}} \tag{15}$$

where sim$(\cdot)$ represents a predefined similarity measure such as the negative of Euclidean distance.

Given a sample set $\mathcal{A}$, we then define the probability of feature $\mathbf{v}$ being recognized as a sample in $\mathcal{A}$ as

$$P(\mathcal{A}|\mathbf{v}) = \sum_{n\in\mathcal{A}} P(n|\mathbf{v}). \tag{16}$$

We then define a confidence score score$_i$ to assess the reliability of the ith superpixel's pseudo-label using (16)

$$\mathrm{score}_i = \frac{P(\mathcal{S}_i \cap \mathcal{C}_i|\mathbf{v}_i)}{P(\mathcal{C}_i|\mathbf{v}_i)}. \tag{17}$$

A lower score$_i$ indicates a higher likelihood of label noise for s_i. To provide further explanation on score$_i$ as a measure of noisy labels, we analyze its conceptual meaning. Let $\mathcal{S}_i^c$ denote the complement of $\mathcal{S}_i$. Then, the probability of class neighbors given spectral feature $\mathbf{v}_i$ can be decomposed as

$$P(\mathcal{C}_i|\mathbf{v}_i) = P\left(\mathcal{S}_i^c \cap \mathcal{C}_i|\mathbf{v}_i\right) + P(\mathcal{S}_i \cap \mathcal{C}_i|\mathbf{v}_i). \tag{18}$$

Thus, from this equation, it is evident that score$_i$ is minimized when $P(\mathcal{S}_i^c \cap \mathcal{C}_i|\mathbf{v_i})$ is maximized and $P(\mathcal{S}_i \cap \mathcal{C}_i|\mathbf{v_i})$ is minimized. This aligns with our objective, as a lower score$_i$ indicates that the pseudo-label is more likely to be noisy, making it a reliable correction criterion. If score$_i$ is smaller than a predefined reliability threshold θ, the pseudo-label p_i is corrected by selecting the next most probable class

$$p_i = \arg\max_{k,k\neq p_i} \mathbf{p}_i. \tag{19}$$

In addition, if a superpixel is labeled (i.e., $l_i \neq -1$), its pseudo-label is directly replaced with its supervised label.

After the supervised correction process, the refined superpixel-level pseudo-labels $(p_i)_{i=1}^N$ are transformed back to the pixel level to generate the final pseudo-label map

$$\overline{\mathbf{P}}(h,w) = p_i, \quad \text{for all } (h,w) \in s_i \tag{20}$$

where $\overline{\mathbf{P}} \in \mathbb{N}^{H\times W}$ is the final pseudo-label map for distillation. The supervised pseudo-label correction algorithm is outlined in Algorithm 1.

Algorithm 1 Pseudo-Code of the Pseudo-Label Correction

Input : Class set Y_U and Y_S, mean features of superpixels $\{\mathbf{v}_i\}_{i=1}^N$, pseudo-label probabilities $(\mathbf{p}_i)_{i=1}^N$, superpixel labels $\{l_i\}_{i=1}^N$, threshold θ
Output: Refined pseudo-labels $\{p_i\}_{i=1}^N$

```
for i = 1 to N do
    p_i ← arg max(p_i);
    if l_i ≠ −1 then
        p_i ← l_i;
    else
        j ← 1;
        while p_i ∉ Y_U do
            Get class neighbors C_i ;
            Get spectral neighbors S_i ;
            Compute reliability score_i:
                score_i = P(S_i ∩ C_i|v_i) / P(C_i|v_i)
            if score_i < θ then
                j ← j + 1 ;
                p_i ← argsort(p_i)[−(j + 1)]
            else
                break
            end
        end
    end
end
```

D. Global Learning for Final Classification

To distill the knowledge of a pretrained VLM using whole pseudo-labels, we primarily use a global classification architecture based on U-Net for HSI dense classification. U-Net is a semantic segmentation model consisting of two main parts: an encoder and a decoder, as shown in Fig. 1. The encoder extracts features from the input image and progressively downsamples it, producing hierarchical low-resolution feature representations. The decoder then uses the abstract features extracted by the encoder to recover the original image size, ultimately generating the segmentation results.

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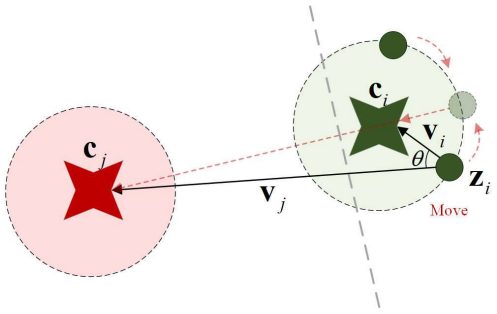


Fig. 5. Illustration of the distribution loss.

Let the model's output prediction be $\mathbf{F} \in \mathbb{R}^{H\times W\times K}$, and let the one-hot label for the training samples be denoted as $\mathbf{Y} \in \mathbb{R}^{H\times W\times K}$, where pixels without labels are represented by zero vectors. The cross-entropy loss for supervised training is defined as follows:

$$\begin{aligned}\mathcal{L}_{\text{sup}} &= \text{CrossEntropy}\left(\mathbf{F}, \mathbf{Y}\right) \\ &= -\frac{1}{\sum_{h=1}^{H}\sum_{w=1}^{W}\mathbf{1}_{\{\mathbf{Y}_{h,w,:}\neq\mathbf{0}\}}}\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{k=1}^{K}Y_{h,w,k}\log F_{h,w,k}.\end{aligned} \tag{21}$$

In addition to the supervised loss, the corresponding pseudo-label cross-entropy loss is defined as

$$\begin{aligned}\mathcal{L}_{\text{pl}} &= \text{CrossEntropy}\left(\mathbf{F}, \overline{\mathbf{P}}\right) \\ &= -\frac{1}{\text{HW}}\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{k=1}^{K}\overline{P}_{h,w,k}\log F_{h,w,k}.\end{aligned} \tag{22}$$

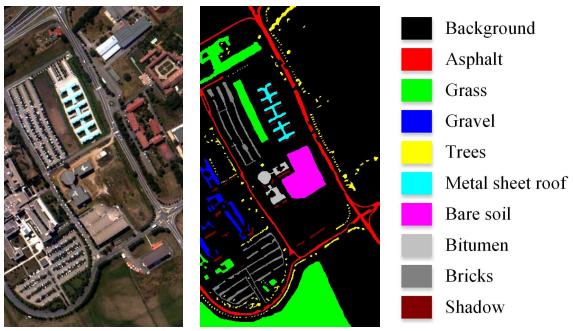
In addition to the two cross-entropy losses, we also apply a distribution loss to constrain the distribution of feature embeddings, encouraging them to align in the reverse direction of nearby negative class centers [88]. This reduces overlap between class clusters and expands the open space, thereby facilitating better discrimination of unknown classes. Specifically, for a feature embedding $\mathbf{z}_i$ of class y_i, the distribution loss is formulated as

$$\mathcal{L}_{\text{dis}} = \frac{1}{M}\sum_{i=1}^{M}\sum_{j\neq y_i}w_j\left(1-\cos\left(\mathbf{v}_i, \mathbf{v}_j\right)\right) \tag{23}$$

where $\mathbf{v}_i = \mathbf{c}_{y_i} - \mathbf{z}_i$ and $\mathbf{v}_j = \mathbf{c}_j - \mathbf{z}_i$ are the vectors from $\mathbf{z}_i$ to the positive and negative class centers, respectively. The weights v_j are defined as

$$w_j = \frac{e^{-D_{ji}}}{\sum_{n\neq y_i}e^{-D_{ni}}} \tag{24}$$

with D_{ji} represents the Euclidean distance between $\mathbf{z}_i$ and the negative class center $\mathbf{c}_j$. As shown in Fig. 5, minimizing (23) drives the angle θ—between the vector from sample feature $\mathbf{z}_i$ to the positive class center $\mathbf{c}_{y_i}$ (simplified as $\mathbf{c}_i$ in the figure) and the vector from $\mathbf{z}_i$ to the negative class center $\mathbf{c}_j$—to approach 0°. Feature embeddings are then preferentially distributed on the opposite side of these negative centers, thereby reducing cluster overlap and expanding the open space between classes. This provides more opportunities for the identification of unknown classes.



(a) (b)

Fig. 6. Pavia University dataset: (a) false-color map and (b) ground-truth.

TABLE I
LAND COVER CLASSES AND NUMBERS OF SAMPLES FOR THE PAVIA UNIVERSITY DATASET

No.	Class Name	Training	Test	Total
1	Asphalt	332	6299	6631
2	Grass	933	17716	18649
3	Gravel	105	1994	2099
4	Trees	154	2910	3064
5	Metal sheet roof	68	1277	1345
6	Soil	252	4777	5029
7	Bitumen	67	1263	1330
8	Bricks	185	3497	3682
9	Shadow	48	899	947
	Total	2144	40632	42776

Finally, the total loss for training the classification model is the weight sum of the above losses

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sup}} + \beta\mathcal{L}_{\text{pl}} + \lambda\mathcal{L}_{\text{dis}} \tag{25}$$

where β and λ are weighting factors. This combined loss ensures accurate classification while optimizing the spatial distribution of feature embeddings for better generalization, especially in generalized zero-shot classification tasks.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Datasets Description

In the experiments, three public HSI datasets are used to evaluate the performance of the proposed method. These datasets are introduced in detail below.

1) Pavia University: The Reflectance Optical System Imaging Spectrometer (ROSIS) sensor acquired this dataset over the University of Pavia, Italy. It consists of 610 × 340 pixels with a spatial resolution of 1.3 × 1.3 m. It also includes 103 spectral bands, whose wavelength ranges from 430 to 860 nm. A total of nine land-covers are labeled in this dataset. Fig. 6 shows a false-color image of the dataset and the corresponding ground-truth map, while Table I lists the numbers of training and test samples for each class.

2) Houston: This dataset describes an urban area surrounding the University of Houston campus in the USA. It was captured by the National Center for Airborne Laser Mapping (NCALM) and then was used for the 2013 IEEE GRSS Data Fusion Contest. It contains 349 × 1905 pixels, with a

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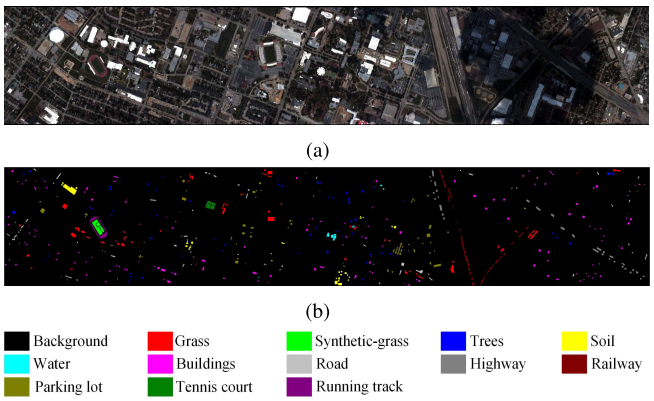


Fig. 7. Merged Houston dataset: (a) false-color map and (b) ground-truth.

TABLE II
LAND COVER CLASSES AND NUMBERS OF SAMPLES FOR THE MERGED HOUSTON DATASET

No.	Class Name	Training	Test	Total
1	Grass	126	2379	2505
2	Synthetic grass	35	662	697
3	Trees	63	1181	1244
4	Soil	63	1179	1242
5	Water	17	308	325
6	Buildings	126	2386	2512
7	Road	63	1189	1252
8	Highway	62	1165	1227
9	Railway	62	1173	1235
10	Parking lot	86	1616	1702
11	Tennis court	22	406	428
12	Running track	33	627	660
	Total	758	14271	15029

spatial resolution of 2.5 ×2.5 m. The dataset encompasses 144 bands with a wavelength range of 380–1050 nm. It contains 15 029 labeled pixels. Originally, 15 different land-cover types were recorded in this dataset. However, there are some land-covers which are hard and meaningless to distinguish, such as parking-lot-1 and parking-lot-2. Therefore, we merge these categories that are not easily distinguishable, and the merged dataset contains 12 classes. Fig. 7 depicts the false-color image and the merged ground-truth map, while Table II provides the number of training and test samples for each class.

3) QUH-Qingyun: In 2021, the Gaiasky mini2-VN imaging spectrometer on an uncrewed aerial vehicle captured this dataset, which mainly describes a residential area in Qingdao, China. It contains 880 × 1360 pixels, with a spatial resolution of 0.15 × 0.15 m. It includes 270 bands with a wavelength range of 400 to 1000 nm. The ground-truth is composed of 42 776 labeled pixels that belong to nine urban land-covers. Fig. 8 displays the false-color image and corresponding ground-truth map of the dataset, while Table III provides the number of training and test samples for each class.

B. Experimental Setup

In the setting of GZSL, it is essential to partition the classes into seen and unseen classes. To achieve this, we select

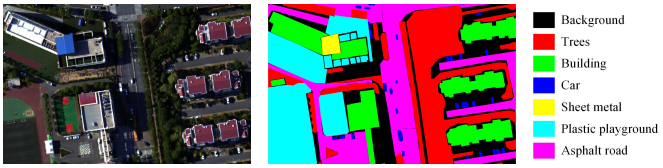


Fig. 8. QUH-Qingyun dataset: (a) false-color map and (b) ground-truth.

TABLE III
LAND COVER CLASSES AND NUMBERS OF SAMPLES FOR THE QUH-QINGYUN DATASET

No.	Class Name	Training	Test	Total
1	Trees	1391	276759	278150
2	Building	898	178614	179512
3	Car	69	13714	13783
4	Sheet metal	49	9718	9767
5	Plastic playground	1089	216646	217735
6	Asphalt road	1280	254666	255946
	Total	4776	950117	954893

a subset of classes from the given hyperspectral dataset as unseen classes, while the remaining classes' training samples are utilized to train the model. For a comprehensive evaluation, the experiment includes the selection of all possible unseen class combinations. Taking the Pavia dataset as an example, which consists of nine classes, selecting two distinct classes as unseen during training results in $C_9^2 = 36$ unique combinations. For each combination, the experiment is conducted over five independent runs, and the AA is reported. Given computational constraints, we primarily focus on the two-unseen-class setting, which provides a representative assessment of the model's performance.

For GZSL evaluation, where the classifier must recognize both seen and unseen HSI pixels in the testing stage, we report the overall accuracy (OA) and AA for both seen and unseen classes, denoted as S(oa), S(aa), U(oa), U(aa), respectively. In addition, we introduce the harmonic mean accuracy (H), which balances the classification performance for seen and unseen classes

$$H = \frac{2}{\frac{1}{S} + \frac{1}{U}} = \frac{2 \times S \times U}{S + U}. \tag{26}$$

Thus, the accuracies H(oa) and H(aa) can be obtained. Also, the traditional evaluation metrics like OA, AA, and Kappa coefficient of the whole testing pixels are also considered.

The two CLIP models used in this research are 1) CLIP-RSICD, which is based on ViT-B/32 fine-tuned using the rsicd remote sensing caption dataset; and 2) CLIP-ViT-L/16, which is pretrained by OpenAI using the natural image–text dataset. The model weights can be obtained at Hugging Face. The architecture of the segmentation network U-Net is illustrated in Fig. 1. For the label correction stage, the spectral clustering method is used for finding spectral neighbors. To train the HZSCM, the Adam optimizer is used and the learning rate is 0.0005 and the total number of training epochs is 600. The

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early stopping strategy is adopted when the loss of pseudo-label learning is less than 0.2 for four consecutive epochs.

Since no existing GZSL methods have been specifically designed for HSI classification, we compare HZSCM with several representative HSI classification methods to validate its effectiveness. These include: local learning-based CNN classification methods, including CNN and DBDA [89], local learning-based transformer methods, including CT-Mixer [90], SSFTT [43], and GSC-ViT [17], semi-supervised learning method DSCA [91], and global learning-based GCN methods, including CNN-GCN [92] and MSSGU [93].

1) *CNN:* It is a basic spectral–spatial model with four 2-D convolutional layers, serving as a foundational baseline for HSI classification.
2) *DBDA:* DBDA is an improved CNN with a double-branch dual-attention mechanism to enhance discriminative feature extraction. It is a representative attention-based CNN baseline, different from transformer-based methods that rely on self-attention.
3) *SSFTT:* It is a hybrid spatial–spectral transformer using Gaussian-weighted tokenization and CNN-derived local features to combine convolutional and transformer strengths.
4) *CT-Mixer:* It is an aggregated framework integrating CNNs into transformer block designs, aiming to synergize the local feature extraction of CNNs with the long-range dependency modeling of transformers.
5) *GSC-ViT:* GSC-ViT is a lightweight variant of the vision transformer that incorporates grouped convolutions and decomposes self-attention into local and global components. It aims at balancing computational efficiency and feature representation capability.
6) *CNN-GCN:* It is a heterogeneous network combining CNN and GCN branches. CNNs learn features from small regular regions, while GCNs model large irregular regions at the superpixel level, generating complementary spectral–spatial features for global classification.
7) *MSSG:* It is a graph-based method utilizing a superpixel-structured graph U-Net to learn multiscale features across hierarchical graphs. It focuses on global spatial relationships, serving as a representative global learning method for comparison with patch-based methods.
8) *DSCA:* It is an explicit CNN-based semi-supervised model. It uses superpixel-guided label propagation and adopts a consistency regularization-based dual-stream network training with dynamic class-adaptive thresholds.
9) *U-Net:* It is a vanilla CNN-based segmentation model with two encoders and two decoder layers, designed for global HSI classification by taking the entire image as input and outputting a pixel-level probability map.

Besides, we have designed GSC-ViT-T and MSSGU-T for fair comparison: they are the modifications of GSC-ViT and MSSGU, respectively. This modification uses the CLIP text encoder to convert class names (both seen and unseen classes) into fixed semantic embeddings, which serve as class prototypes. And then the similarity between extracted image features and semantic prototypes is computed, and their total loss is updated to a weighted sum of their original classification loss and a prototype similarity loss. This modification ensures the baseline methods can equally access class name information.

Generally, the comparison methods cover different but sufficient HSI classification methods for comparison.

C. Classification Results

To comprehensively evaluate the effectiveness of the proposed GZSL method for HSI classification, the experimental results are presented in three parts. Under the setting where two classes are designated as unseen, the first part provides an overall comparison of the mean classification accuracies achieved by the proposed method and baseline approaches across all possible unseen class combinations. The second part further analyzes the classification performance of the proposed method for each specific unseen class combination, highlighting the variability across different scenarios. Finally, the third part presents per-class classification accuracies for selected unseen class combinations.

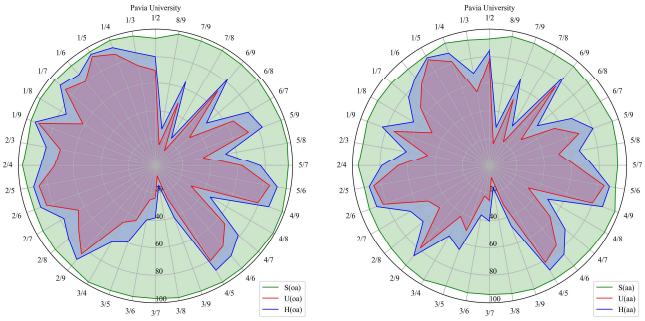
1) General Classification Results: Table IV shows the mean generalized zero-shot classification results in the case of two unseen classes. From the results, it can be seen that the proposed HZSCM is effective for the HSI GZSL task. Although there are unseen classes in the test stage, the proposed method achieves considerable classification accuracy not only for the seen classes but also for the unseen classes. Specifically, it can be seen the following aspects.

1) All the outlined classification methods have all obtained high accuracies for the seen classes. Generally speaking, the global learning-based classification methods obtain higher results than local learning-based methods in the setting. For example, the MSSGU outperforms SSFTT by 0.58 percentage points, 0.73 percentage points in terms of S(oa) and S(aa) on the Pavia University dataset.
2) The traditional HSI classification, including semi-supervised learning methods, cannot recognize unseen classes in the ZSL task. The accuracies of unseen classes U(oa) and U(aa) are all zeros for all the comparison methods in the three datasets, and therefore their harmonic mean accuracies are all zeros too.
3) Both GSC-ViT-T and MSSGU-T exhibit lower classification seen class accuracies than their original counterparts (GSC-ViT and MSSGU) across all datasets. We infer that the underperformance of GSC-ViT-T and MSSGU-T stems from the inherent limitation of using category names as prototypes: semantic embeddings of similar category names are located close in the feature space, which reduces interclass discriminability and increases classification difficulty. Besides, despite accessing category name information, neither GSC-ViT-T nor MSSGU-T gains the ability to recognize unseen classes—their U(oa) and U(aa) remain 0% across all datasets.
4) The HZSCM model demonstrates a strong capability in recognizing unseen classes while maintaining high seen-class accuracy. For example, on the QUH-Qingyun dataset, HZSCM achieves a mean unseen OA of 69.63%

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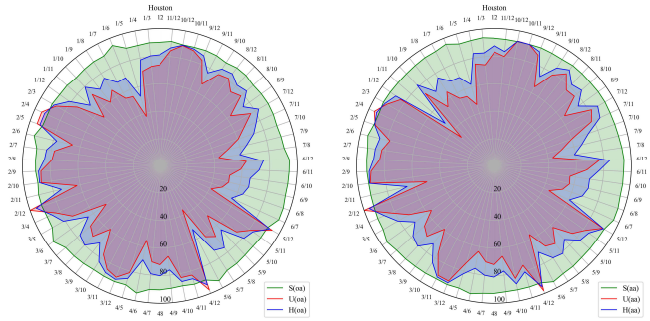
TABLE IV
GENERAL TEST DATA CLASSIFICATION RESULTS ON THE THREE HSI DATASETS

		CNN	DBDA	SS-FTT	CT-Mixer	GSC-ViT	GSC-ViT-T	CNN-GCN	MSSGU	MSSGU-T	DSCA	U-Net	HZSCM
Pavia University	S(oa) (%)	99.07	99.21	99.33	99.04	99.70	99.48	99.90	99.91	99.62	99.58	99.75	96.79
	U(oa) (%)	0	0	0	0	0	0	0	0	0	0	0	60.79
	H(oa) (%)	0	0	0	0	0	0	0	0	0	0	0	71.17
	S(aa) (%)	98.96	98.94	99.12	98.89	99.57	99.02	99.83	99.85	99.63	99.32	99.49	94.52
	U(aa) (%)	0	0	0	0	0	0	0	0	0	0	0	57.16
	H(aa) (%)	0	0	0	0	0	0	0	0	0	0	0	68.38
Houston	S(oa) (%)	97.37	96.74	97.11	97.86	97.41	97.03	98.54	98.41	98.23	98.25	97.64	90.96
	U(oa) (%)	0	0	0	0	0	0	0	0	0	0	0	70.50
	H(oa) (%)	0	0	0	0	0	0	0	0	0	0	0	78.28
	S(aa) (%)	97.46	96.91	97.22	98.03	97.58	97.18	98.80	98.63	98.31	98.31	97.89	93.20
	U(aa) (%)	0	0	0	0	0	0	0	0	0	0	0	72.66
	H(aa) (%)	0	0	0	0	0	0	0	0	0	0	0	80.49
QUH-Qingyun	S(oa) (%)	96.20	96.87	95.42	96.48	97.11	96.75	97.95	97.85	97.58	97.63	98.94	96.17
	U(oa) (%)	0	0	0	0	0	0	0	0	0	0	0	69.63
	H(oa) (%)	0	0	0	0	0	0	0	0	0	0	0	78.92
	S(aa) (%)	90.77	92.52	89.56	91.47	93.41	93.02	95.40	96.24	95.88	94.82	94.89	93.55
	U(aa) (%)	0	0	0	0	0	0	0	0	0	0	0	74.26
	H(aa) (%)	0	0	0	0	0	0	0	0	0	0	0	82.17



(a) (b)

Fig. 9. Mean seen, unseen, and harmonic accuracies on the Pavia University dataset: (a) overall accuracies and (b) average accuracies.

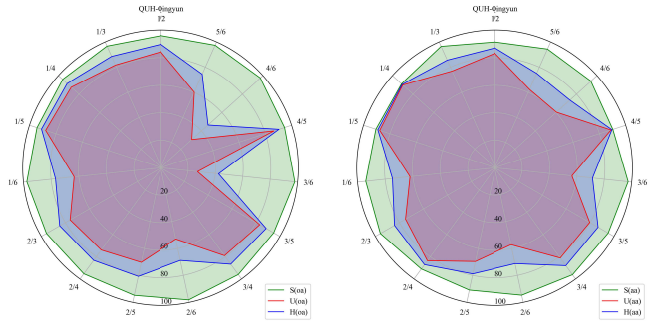


(a) (b)

Fig. 10. Mean seen, unseen, and harmonic accuracies on the merged Houston dataset: (a) overall accuracies and (b) average accuracies.

and unseen AA of 74.26%, showcasing its effectiveness in generalizing to novel classes. However, incorporating zero-shot recognition slightly impacts seen-class performance, as evidenced by a minor decrease in S(oa) = 96.17% and S(aa) = 93.55% compared to other methods. Similar conclusions can also be drawn from the other two datasets. Nevertheless, the proposed HZSCM still maintains good classification ability for seen classes.

2) Specific GZSL Classification Performance for All Combinations of Unseen Classes: Figs. 9–11 illustrate the classification performance of the proposed GZSL model across different combinations of unseen classes for the three datasets. The results demonstrate that the proposed method maintains a reasonable zero-shot classification capability while preserving high classification accuracy for seen classes. However, the classification performance varies depending on the specific combination of unseen categories. For certain class combinations, the model exhibits relatively poor performance. For example, on the Pavia University dataset, when the third category, gravel, is designated as an unseen class, the model



(a) (b)

Fig. 11. Mean seen, unseen, and harmonic accuracies on the QUH-Qingyun dataset: (a) overall accuracies and (b) average accuracies.

struggles to recognize it correctly. A similar issue is observed for the eighth category, bricks. This may be attributed to the fact that both categories represent surface materials commonly found in road construction, which are not only semantically uncommon but also lack distinctive spatial characteristics,

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TABLE V

TESTING DATA CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE PAVIA UNIVERSITY DATASET

	(1,2)*			(5,8)			(3,7)		
	CT-Mixer	MSSGU	HZSCM	CT-Mixer	MSSGU	HZSCM	CT-Mixer	MSSGU	HZSCM
	0.00±0.00	0.00±0.00	96.29±0.80	99.97±0.03	100.0±0.00	99.11±0.50	99.97±0.05	100.0±0.00	99.11±0.57
	0.00±0.00	0.00±0.00	60.65±0.49	99.99±0.02	100.0±0.00	99.77±0.17	98.58±2.81	100.0±0.00	99.25±0.18
	93.61±10.98	100.0±0.00	93.90±0.77	99.55±0.17	100.0±0.00	93.48±1.12	0.00±0.00	0.00±0.00	15.70±1.25
	98.74±0.28	99.21±0.36	81.94±1.42	98.49±0.14	99.13±0.47	83.90±2.46	98.87±0.41	98.36±0.89	83.90±2.32
	99.80±0.16	100.0±0.00	99.67±0.40	0.00±0.00	0.00±0.00	85.59±0.31	99.75±0.15	100.0±0.00	99.44±0.57
	99.82±0.25	100.0±0.00	99.61±0.54	99.64±0.63	100.0±0.00	99.48±0.56	92.69±14.22	100.0±0.00	98.92±1.01
	99.89±0.22	100.0±0.00	98.46±0.82	99.72±0.53	100.0±0.00	98.27±0.99	0.00±0.00	0.00±0.00	36.33±0.15
	99.34±0.46	100.0±0.00	91.88±1.24	0.00±0.00	0.00±0.00	17.27±2.82	99.70±0.32	99.99±0.01	91.31±1.05
	98.69±0.95	99.93±0.05	84.49±2.10	99.82±1.38	99.98±0.04	85.45±3.31	99.00±0.69	99.96±0.05	86.00±4.13
OA (%)	39.15±0.58	39.60±0.03	**79.24±0.28**	87.49±0.06	87.64±0.03	**89.94±0.45**	89.98±1.82	**91.48±0.06**	90.78±0.07
AA (%)	76.65±1.28	77.68±0.04	**89.66±0.29**	77.35±0.11	77.68±0.05	**84.70±0.93**	76.51±1.60	**77.59±0.09**	78.89±0.53
K×100	33.99±0.60	34.56±0.17	**73.93±0.35**	83.49±0.12	83.55±0.09	**86.64±0.62**	86.62±2.49	**88.62±0.08**	87.76±0.11
S(aa) (%)	98.55±1.64	**99.88±0.05**	92.85±0.60	99.45±0.14	**99.87±0.07**	94.21.±0.88	98.37±2.05	**99.76±0.12**	93.99±0.80
U(aa) (%)	0.00±0.00	0.00±0.00	**78.47±0.41**	0.00±0.00	0.00±0.00	**51.43±1.42**	0.00±0.00	0.00±0.00	**26.02±0.57**
H(aa) (%)	0.00±0.00	0.00±0.00	**85.06±0.30**	0.00±0.00	0.00±0.00	**66.53±1.35**	0.00±0.00	0.00±0.00	**40.75±0.65**

*(1,2) means that the first and second classes are unseen during training, and the same applies thereafter.

TABLE VI

TESTING DATA CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE MERGED HOUSTON DATASET

	(3,11)			(1,11)			(7,8)		
	CT-Mixer	MSSGU	HZSCM	CT-Mixer	MSSGU	HZSCM	CT-Mixer	MSSGU	HZSCM
	99.97±0.04	99.73±0.54	95.51±1.90	0.00±0.00	0.00±0.00	63.63±6.02	99.22±0.55	99.48±0.61	95.71±1.47
	99.34±0.47	99.85±0.17	100.0±0.00	99.34±0.45	99.76±0.20	100.0±0.00	99.43±0.43	99.82±0.29	100.0±0.00
	0.00±0.00	0.00±0.00	79.68±1.58	99.15±0.46	99.98±0.03	88.42±2.32	97.97±2.05	100.0±0.00	89.31±3.03
	99.98±0.03	99.92±0.17	97.03±2.56	99.95±0.10	99.98±0.03	98.88±1.99	99.86±0.13	99.92±0.17	97.22±2.40
	95.78±4.38	98.51±0.81	93.44±2.39	95.84±4.41	97.14±4.14	97.79±0.38	95.84±4.41	98.51±0.81	92.40±3.42
	95.44±2.16	97.82±1.31	72.98±2.38	94.82±1.25	97.02±1.77	73.81±3.02	96.35±1.95	97.74±1.38	74.04±3.29
	96.23±2.23	92.03±3.36	85.00±2.48	96.10±1.29	92.13±5.63	84.41±3.28	0.00±0.00	0.00±0.00	58.18±9.03
	99.85±0.31	99.79±0.41	89.99±4.12	100.0±0.00	98.88±1.34	90.56±3.73	0.00±0.00	0.00±0.00	28.56±0.89
	96.97±1.67	100.0±0.00	97.87±1.57	98.07±1.28	99.59±0.82	97.86±1.67	98.01±1.46	99.61±0.78	97.61±1.53
	96.56±2.78	97.38±1.96	94.79±1.46	98.50±0.40	96.39±3.44	95.16±1.69	99.38±0.47	98.19±0.89	94.31±1.63
	0.00±0.00	0.00±0.00	100.0±0.00	0.00±0.00	0.00±0.00	0.70±0.49	100.0±0.00	100.0±0.00	99.85±0.20
	100.0±0.00	99.97±0.06	99.97±0.06	99.94±0.12	100.0±0.00	99.94±0.00	100.0±0.00	99.97±0.06	100.0±0.00
OA (%)	86.51±0.60	86.93±0.23	**89.80±0.68**	77.96±0.24	77.91±0.60	**82.55±1.26**	81.52±0.38	82.04±0.21	**83.16±0.59**
AA (%)	81.68±0.23	82.08±0.29	**92.19±0.40**	81.81±0.30	81.74±0.82	**82.60±0.57**	82.17±0.24	82.77±0.15	**85.60±0.70**
K×100	84.72±0.67	85.19±0.26	**88.61±0.76**	75.69±0.26	75.56±0.69	**80.47±1.39**	79.18±0.41	79.65±0.24	**81.17±0.67**
S(aa) (%)	98.01±0.28	**98.50±3.50**	92.66±0.33	**98.17±0.37**	98.09±0.99	92.69±0.57	98.61±0.29	**99.32±0.18**	94.05±0.55
U(aa) (%)	0.00±0.00	0.00±0.00	**89.84±0.79**	0.00±0.00	0.00±0.00	**32.17±3.06**	0.00±0.00	0.00±0.00	**43.37±4.23**
H(aa) (%)	0.00±0.00	0.00±0.00	**91.23±0.55**	0.00±0.00	0.00±0.00	**47.67±3.37**	0.00±0.00	0.00±0.00	**59.24±3.81**

making them prone to confusion. This challenge is evident in the RGB imagery of Pavia University, where these land-cover types exhibit substantial spectral and spatial similarities. A similar pattern can be observed on the Houston dataset. The model performs well on the QUH-Qingyun dataset, where it achieves consistently high classification accuracy for both seen and unseen classes across most class combinations. However, when the third and sixth categories are set as unseen classes, the OA of the unseen classes drops to approximately 30%, while the AA remains around 60%. This may be due to the strong correlation between road and asphalt road, as these two land-cover types frequently co-occur, making their discrimination more challenging. These results suggest that the model's performance is influenced by HSI resolution, category granularity, and alignment with CLIP's training data. Higher spatial resolution enhances pseudo-label accuracy, improving zero-shot classification. Conversely, as category definitions become less conventional and interclass entanglement increases, pseudo-label accuracy declines, negatively impacting overall performance.

3) Class-Wise Accuracies for Specific Combinations of Unseen Classes: Tables V–VII present classification results for selected unseen class combinations on the three datasets, covering well-performing and challenging cases. The S, U, and H accuracies are reported as averages.

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TABLE VII
TESTING DATA CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE QUH-QINGYUN DATASET

	(1,4)			(5,6)			(3,6)		
	CT-Mixer	MSSGU	HZSCM	CT-Mixer	MSSGU	HZSCM	CT-Mixer	MSSGU	HZSCM
	0.00±0.00	0.00±0.00	87.41±0.97	99.20±0.17	99.70±0.11	98.92±0.27	98.74±0.29	99.51±0.18	98.18±0.47
	95.34±1.11	99.67±0.09	96.84±0.92	98.13±0.82	99.69±0.04	95.89±0.78	98.37±0.50	99.62±0.11	97.32±0.49
	55.27±1.52	82.76±4.07	75.35±1.96	66.13±8.42	97.23±0.84	86.51±3.25	0.00±0.00	0.00±0.00	89.17±1.22
	0.00±0.00	0.00±0.00	92.66±0.97	97.69±1.11	97.10±1.20	96.02±1.41	97.92±0.73	96.74±0.98	95.86±1.28
	97.81±0.57	99.17±0.23	98.16±0.13	0.00±0.00	0.00±0.00	89.34±0.68	96.72±0.56	99.29±0.09	98.34±0.27
	97.16±1.04	98.52±0.43	94.42±0.97	0.00±0.00	0.00±0.00	35.39±2.36	0.00±0.00	0.00±0.00	23.66±1.20
OA (%)	66.97±0.25	68.85±0.02	**93.38±0.28**	49.17±0.08	50.05±0.03	**78.89±0.79**	70.21±0.06	71.24±0.05	**77.85±0.38**
AA (%)	57.60±0.06	63.35±0.75	**90.81±0.72**	60.19±1.33	65.62±0.20	**83.68±0.75**	65.29±0.16	65.86±0.16	**83.76±0.30**
K×100	56.82±0.18	59.30±0.09	**91.30±0.36**	34.28±0.42	38.57±0.80	**72.61±0.93**	61.15±0.08	62.22±0.08	**71.87±0.45**
S(aa) (%)	86.40±0.10	**95.03±0.11**	91.19±0.77	90.29±0.99	**98.43±0.31**	94.34.±0.64	97.94±0.14	**98.79±0.24**	97.43±0.31
U(aa) (%)	0.00±0.00	0.00±0.00	**90.03±1.19**	0.00±0.00	0.00±0.00	**62.37±1.39**	0.00±0.00	0.00±0.00	**56.41±0.98**
H(aa) (%)	0.00±0.00	0.00±0.00	**90.60±0.79**	0.00±0.00	0.00±0.00	**75.08±1.11**	0.00±0.00	0.00±0.00	**71.45±0.76**

From the results showed in Tables V–VII, it is evident that traditional methods fail to recognize unseen classes due to the absence of a GZSL mechanism or external auxiliary knowledge, resulting in zero accuracy for these classes. In contrast, the proposed method exhibits varying performance across different unseen class combinations, with several noteworthy cases.

On the Pavia University dataset, the model achieves higher zero-shot accuracies when asphalt (Class 1) and grass (Class 2) are the unknown classes. These land covers exhibit minimal spatial entanglement with other classes and are common, allowing CLIP to generate high-quality pseudo-labels, thereby improving classification accuracy. However, for the combination of metal sheet roof (Class 5) and bricks (Class 8), the model performs well on metal sheet roof but poorly on bricks. This discrepancy likely arises because the metal sheet roof aligns more closely with CLIP's prior knowledge, whereas bricks lack distinct visual cues in this remote sensing image, making them difficult to recognize. Similarly, the model struggles with gravel (Class 3) and bitumen (Class 7). Examination of the original image suggests that gravel refers to man-made structures rather than loose gravel, while bitumen primarily represents asphalt roofs—both of which deviate from conventional CLIP concepts, leading to misclassification.

On the merged Houston dataset, when grass (Class 1) and tennis court (Class 11) are both unseen, the model struggles to classify tennis court, with nearly zero accuracy. In contrast, when gravel (Class 3) and tennis court (Class 11) are the unseen classes, the latter achieves 100% accuracy. This discrepancy likely stems from the visual similarities between grass and tennis court, as their green appearances in the dataset lead to frequent misclassification. However, when grass is absent from the unseen set, spectral-based pseudo-label correction mechanisms help distinguish these classes using their distinct spectral features. In addition, when road (Class 7) and highway (Class 8) are unseen, their accuracies are relatively low, likely due to their high similarity in both spectral features and class semantics, making them difficult to discriminate.

The QUH-Qingyun dataset, with its higher spatial resolution and regular class names, yields accurate pseudo-labels and achieves better performance on unseen classes. However, challenges remain. For instance, asphalt road (Class 6) often occupies larger areas and is fragmented by segmentation algorithms, leading to a loss of spatial context and reduced CLIP recognition accuracy. As shown in Table VII, when asphalt road appears as an unseen class in combinations such as (Classes 5 and 6) and (Classes 3 and 6), its classification accuracy remains relatively low.

From the results and analysis of the above, we can make several observations: 1) the higher the spatial resolution of the HSI, the better the recognition performance of the model in general; 2) the category labeling of the HSI should align with standard practices, as this improves the model's the recognition performance; and 3) for spatially indistinguishable land-covers, the inclusion of additional expert knowledge such as spectral information will further improve the GZSL capability of the model.

D. Ablation Study and Parameter Analysis

The proposed method consists of multiple key components, necessitating ablation studies and parameter analysis to assess their individual contributions. First, we evaluate the effects of multimodel ensembling and SAM-based enhancement, as shown in Table VIII, which reports the accuracy of pseudo-labels under different module configurations. The results highlight that each module positively impacts classification performance.

Starting with the baseline configuration using only the CLIP-ViT module (V^* active), the model offers the lowest performance across all metrics (OA, AA, and K) on the Pavia University, Merged Houston, and QUH-Qingyun datasets, with OA values of 42.84%, 32.24%, and 33.67%, respectively. This confirms that relying solely on CLIP-ViT is insufficient for effective HSI classification.

When integrating the CLIP-RSICD module (R^*), classification performance improves significantly, yielding OA values of 49.16%, 37.08%, and 39.57% across the three

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TABLE VIII
ABLATION EXPERIMENTS FOR DIFFERENT MODULES

V*	R*	S*	Pseudo-Label's Accuracies	Pavia University	Merged Houston	QUH-Qingyun
✓	×	×	OA (%)	42.84	32.24	33.67
			AA (%)	30.75	31.18	48.15
			K×100	30.06	24.57	23.05
×	✓	×	OA (%)	38.37	28.61	42.08
			AA (%)	24.38	35.70	52.97
			K×100	24.34	22.79	31.05
✓	✓	×	OA (%)	49.16	37.08	39.57
			AA (%)	32.65	37.56	53.56
			K×100	36.85	31.07	29.03
✓	×	✓	OA (%)	51.36	32.56	48.26
			AA (%)	39.84	31.14	59.12
			K×100	38.89	24.84	35.75
×	✓	✓	OA (%)	39.37	38.69	59.16
			AA (%)	24.36	51.99	48.96
			K×100	25.55	33.67	49.47
✓	✓	✓	OA (%)	51.68	43.17	61.11
			AA (%)	40.49	50.46	68.20
			K×100	39.38	37.83	51.45

* V: CLIP-ViT, R: CLIP-RSICD, S: SAM.

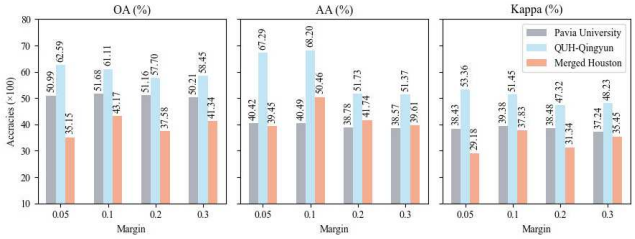


Fig. 12. Accuracies of unsupervised pseudo-labels with different values of margin.

datasets. This demonstrates the positive impact of incorporating domain-specific remote sensing knowledge. Furthermore, adding the SAM module (S*) further enhances performance, with the full model configuration (*V**, *R**, and *S**) achieving the highest OA of 51.68%, 43.17%, and 61.11%, respectively. These results validate the hypothesis that the synergy among CLIP-ViT, CLIP-RSICD, and SAM modules leads to superior classification accuracy. Notably, removing CLIP-ViT (disabling V* while retaining *R** and *S**) results in a substantial drop in OA, with Pavia University decreasing to 39.37%, underscoring the critical role of CLIP-ViT. Overall, the findings highlight that integrating all three modules maximizes classification accuracy, demonstrating the efficacy of each component for the final classification.

Fig. 12 illustrates the accuracy of unsupervised pseudo-labels with varying margin values for the three datasets. The results indicate a clear performance dependency on margin selection. The highest OA for Pavia University (51.68%) is observed at a margin of 0.1, with lower performance at margins of 0.05 and 0.3 (50.99% and 51.11%, respectively).

TABLE IX
ABLATION EXPERIMENT FOR DIFFERENT RGB-INPUT GENERATION METHODS

RGB Gen. Methods	Pseudo-Label's Accuracies	Pavia University	Merged-Houston	QUH-Qingyun
Naive	OA (%)	51.68	43.17	61.11
	AA (%)	40.49	50.46	68.20
	K×100	39.38	37.83	51.45
Average	OA (%)	46.69	30.66	54.60
	AA (%)	30.92	35.29	65.95
	K×100	33.72	24.22	46.12
Gaussian Weighting	OA (%)	53.38	38.90	59.34
	AA (%)	41.65	38.46	68.55
	K×100	41.08	33.06	50.05

Similarly, the optimal margins for the QUH-Qingyun and Merged Houston datasets are also 0.1, achieving OAs of 43.71% and 37.58%, respectively. These findings suggest that a margin of 0.1 is generally suitable for most datasets.

There are many ways to convert HSIs into RGB images, and different conversion methods should affect the quality of pseudo-labeling. We then conduct an ablation experiment to compare different conversion methods: 1) single-band selection (naive): directly selecting three bands corresponding to the central wavelengths of R (620–750 nm), G (500–565 nm), and B (400–450 nm); 2) range averaging: computes the mean value of all bands within the typical wavelength ranges of R, G, and B to generate each color channel; and 3) Gaussian weighting: the weights are assigned to bands within each color range using a Gaussian function (peaked at the central wavelength) and then sums the weighted bands to generate RGB channels.

The results are shown in Table IX. From these results, we can see that methods with narrower effective bandwidths (i.e., single-band selection and Gaussian weighting) generally outperform range averaging. This suggests that retaining relatively concentrated spectral information around the central wavelengths of R/G/B is more conducive to VLMs' land-cover recognition. The single-band selection method is preferred as it balances simplicity, efficiency, and sufficient pseudo-label quality.

As for the fusion strategies of the dual CLIP models' results, we have supplemented ablation experiments comparing simple averaging with different fusion strategies apart from simply average, with results presented in Table X. 1) Entropy weighting: for each CLIP model's output probability distribution, we calculate entropy (a metric of prediction uncertainty; lower entropy indicates higher reliability) and assign weights inversely proportional to entropy (normalized to sum to 1), prioritizing contributions from more confident predictions. 2) Max-prob weighting: uses the maximum probability value of each model's output as its weight, emphasizing predictions with stronger class certainty. 3) Product fusion: computes the product of dual CLIP outputs (instead of summation) to amplify consensus between high-confidence predictions.

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TABLE X
ABLATION EXPERIMENTS FOR DIFFERENT CLIPS ' RESULTS FUSION METHODS

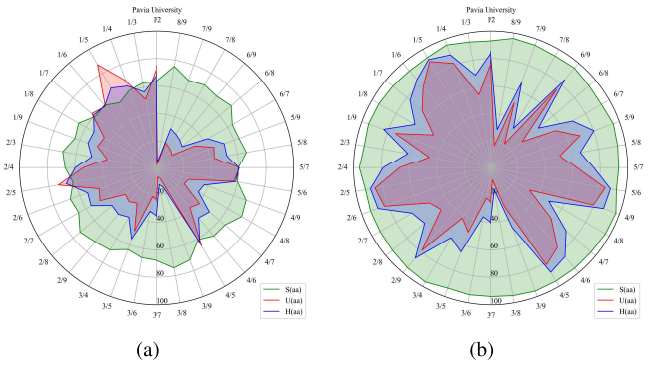
Fusion Methods	Pseudo-Label's Accuracies	Pavia University	Merged-Houston	QUH-Qingyun
Average	OA (%)	51.68	43.17	61.11
	AA (%)	40.49	50.46	68.20
	K×100	39.38	37.83	51.45
Entropy	OA (%)	51.68	40.12	60.84
	AA (%)	40.49	47.18	68.44
	K×100	39.34	34.54	51.05
Max-prob Weighting	OA (%)	51.40	39.68	60.52
	AA (%)	40.20	46.97	67.72
	K×100	39.11	34.05	50.74
Product	OA (%)	51.55	42.82	58.83
	AA (%)	39.04	46.78	66.67
	K×100	38.81	37.33	48.79

From Table X, one can see that simple averaging outperforms alternatives across most metrics. We note that while more sophisticated fusion strategies could potentially further refine pseudo-labels, they are not critical to validating our framework's core idea—i.e., bridging the task paradigm and knowledge gaps between VLMs and HSI classification. Such optimizations can be explored in future work without altering the fundamental GZSL pipeline, but the simple average is indeed simple but effective.

Next, we assess the label correction module through ablation experiments, with the results summarized in Table XI. Incorporating label correction (w) consistently improves classification performance compared to the no-correction scenario (w/o). For example, on the Pavia University dataset, U(oa) increases from 49.20% to 60.79%, while H(oa) rises from 59.52% to 71.17%. Similar trends are observed on the Merged Houston dataset, where U(oa) improves from 29.77% to 70.50%, and H(oa) increases from 41.16% to 78.28%. The QUH-Qingyun dataset also benefits, with U(oa) reaching 69.63% and H(oa) slightly improving to 78.92%. These results confirm that the label correction module enhances generalization to unseen classes.

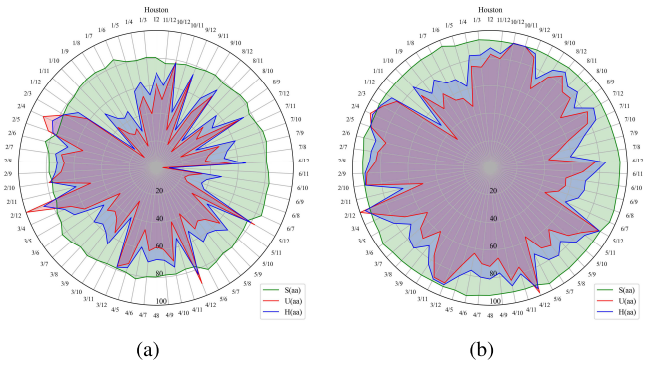
Besides, Figs. 13–15 compare classification results with and without label correction across all unseen class combinations. From these figures, one can see that the zero-shot classification performance is improved by label correction under many unseen class combinations. On the one hand, for certain class combinations, the accuracy of previously misclassified unseen categories improves significantly after label correction. On the other hand, in some cases where zero-shot classification initially fails, label correction enables the model to assign reasonable predictions to unseen samples. These findings demonstrate the effectiveness of label correction in improving zero-shot classification performance.

Finally, the loss weights β and λ are briefly analyzed. Fig. 16, depicts the accuracies [S(aa), U(aa), and H(aa)] for the datasets with varying values of β.In general, a larger β value means that the model will pay more attention to pseudo labels.



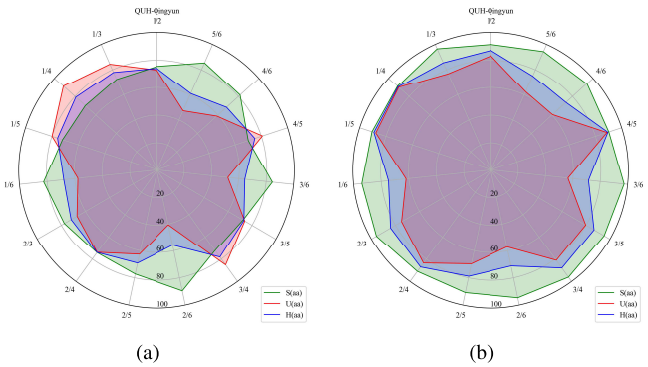
(a) (b)

Fig. 13. Mean seen, unseen, and harmonic average accuracies on the Pavia University dataset: (a) without label correction and (b) with correction.



(a) (b)

Fig. 14. Mean seen, unseen, and harmonic average accuracies on the merged Houston dataset: (a) without label correction and (b) with correction.



(a) (b)

Fig. 15. Mean seen, unseen, and harmonic average accuracies on the QUH-Qingyun dataset: (a) without label correction and (b) with correction.

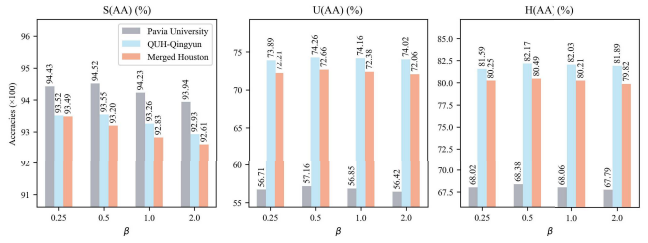


Fig. 16. Accuracies with different values of loss weight β.

If the value of β is too large, the model will overfit to possible noisy labels and ignore the results of supervised training.

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TABLE XI
ABLATION EXPERIMENT FOR LABEL CORRECTION

Dataset	Correction	S(oa) (%)	U(oa) (%)	H(oa) (%)	S(aa) (%)	U(aa) (%)	H(aa) (%)	OA (%)	AA (%)	K×100
Pavia University	w/o	75.70	44.41	51.66	66.19	40.54	46.33	70.46	60.49	61.88
	w	96.79	60.79	71.17	94.52	57.16	68.38	90.35	86.22	87.34
Merged Houston	w/o	74.29	47.74	55.32	80.04	50.58	59.17	68.82	75.14	65.58
	w	90.96	70.50	78.28	93.20	72.66	80.49	86.84	89.78	85.32
QUH-Qingyun	w/o	72.45	62.51	63.79	77.16	68.55	71.06	68.69	74.29	60.51
	w	96.17	69.63	78.92	93.55	74.26	82.17	87.04	87.12	83.17

TABLE XII
TIME ANALYSIS ON THE THREE HSI DATASETS

Dataset	ERS	SAM	CLIP-RSICD	CLIP-ViT-L	Label Correction	U-Net test time.	U-Net FLOPs	U-Net Para.
Pavia University	0.60s	13.29s	18.89s	24.18s	5.98s	4.01ms	89.78G	1.10M
Merged Houston	2.68s	87.62s	33.78s	46.82s	15.68s	6.12ms	289.77G	1.10M
QUH-Qingyun	6.25s	14.63s	21.25s	25.92s	30.65s	9.43ms	523.30G	1.10M

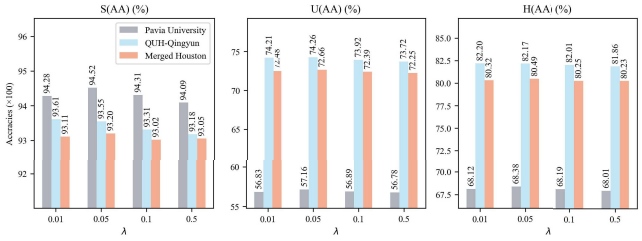


Fig. 17. Accuracies with different values of loss weight λ.

If the β is too small, the model may fail to learn enough about unknown classes, resulting in reduced performance. Values of 0.5 or 1.0 are good choices for many evaluation criteria. Fig. 17 shows the accuracies [S(aa), U(aa), and H(aa)] for the datasets of varying values of λ. Overall, a λ with a value of 0.05 appears to be suitable for balancing accuracy across the different metrics evaluated.

E. Algorithm Complexity

The proposed HZSCM contains several key steps, and it is better to conduct time complexity analysis individually. Table XII shows the test time for the key component of the HZSCM framework, with quantitative results on three benchmark datasets presented in Table R6 (using GPU for deep models like CLIPs, SAM, and U-Net). The analysis focuses on inference-phase time and key preprocessing steps, as follows.

1) *Spatial Segmentation:* Superpixel extraction (ERSs) and SAM-based segmentation introduce modest overhead. Runtime is governed by i) the spatial resolution of the input HSI (larger images increase segmentation time) and ii) the requested number of superpixels (more superpixels raise ERS complexity). SAM segmentation on the Merged Houston dataset is an outlier (87.62 s) because the extreme height-weight ratio (349 × 1905 pixels) forces the image to be split into multiple overlapping tiles to preserve segmentation quality; the redundant inference on overlapping regions dominates the cost.
2) *Pseudo-Label Generation:* Dual-CLIP inference (CLIP-RSICD and CLIP-ViT-L) is the second cost driver. Execution time scales linearly with i) the number of superpixel patches fed to the models and ii) the backbone size: ViT-L (CLIP-ViT-L) is noticeably slower than ViT-B (CLIP-RSICD).
3) *Pseudo-Label Correction:* The spectral information-guided label correction process involves calculating the pairwise cosine similarity between superpixel average spectra. In addition, it requires identifying and correcting pseudo-labels for each superpixel's average spectral vector. The computational time of this stage is largely dependent on the number of superpixels.
4) *U-Net Inference:* Despite high FLOPs, the U-Net delivers a short test time for the entire HSI input: 4.01 ms (Pavia U.), 6.12 ms (Merged Houston), 9.43 ms (QUH-Qingyun). The apparent FLOP overhead is an artifact of the end-to-end paradigm: the network ingests the entire HSI input once and outputs a dense prediction map, whereas patch-based methods repeatedly slide a window across the image but need longer test time for all pixels. The parameter count is fixed and small across all datasets.

F. Classification Maps

Figs. 18–20 show the classification maps for different methods on the three datasets. From these qualitative results, it is evident that the proposed method demonstrates superior classification performance under zero-shot conditions. For instance, on the Pavia University dataset, when the second class, grass, is designated as an unseen class, traditional methods that lack

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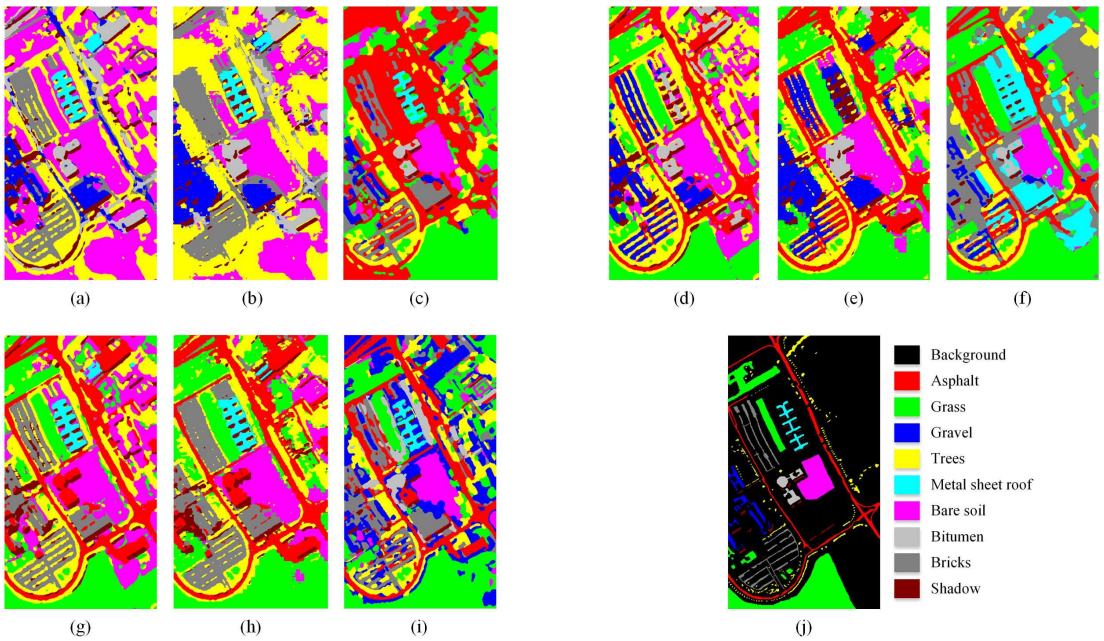


Fig. 18. Classification maps using different methods on the Pavia University dataset. (a)–(c) (First and second classes are unseen) CTMixer, MSSGU, and HZSCM; (d)–(f) (fifth and eighth classes are unseen) CTMixer, MSSGU, and HZSCM; (g)–(i) (third and seventh classes are unseen) CTMixer, MSSGU, and HZSCMT; and (j) ground-truth.

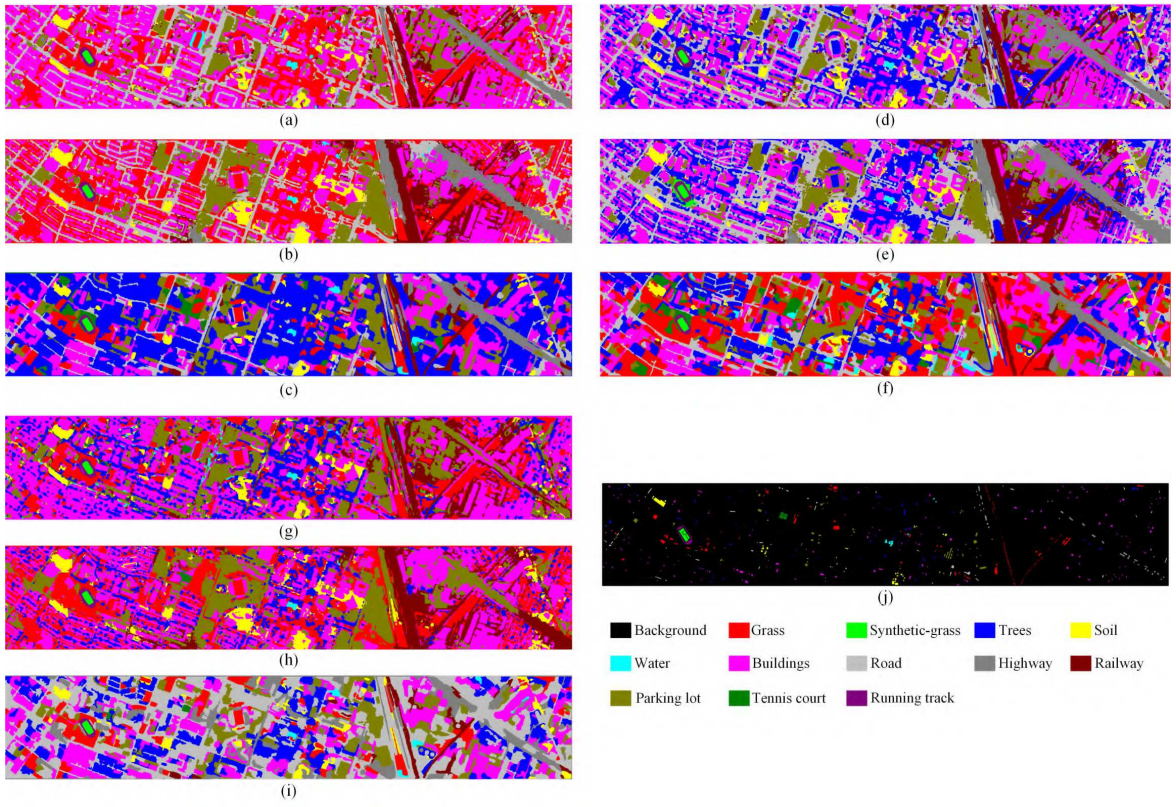


Fig. 19. Classification maps using different methods on the merged Houston dataset. (a)–(c) (Third and eleventh classes are unseen) CTMixer, MSSGU, and HZSCM; (d)–(f) (first and eleventh classes are unseen) CTMixer, MSSGU, and HZSCM; (g)–(i) (seventh and eighth classes are unseen) CTMixer, MSSGU, and HZSCMT; and (j) ground-truth.

zero-shot classification capability tend to misclassify it as trees or bare soil. This misclassification occurs due to the high spectral and spatial similarity between grassland, trees, and bare soil. In contrast, the proposed method effectively distinguishes grassland from these categories, as the VLM provides relevant prior knowledge. This suggests that our approach

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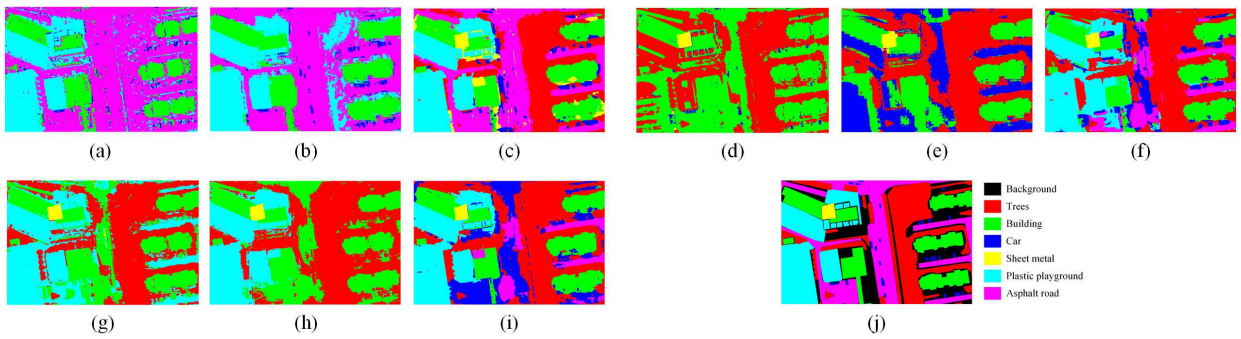


Fig. 20. Classification maps using different methods on the QUH-Qingyun dataset. (a)–(c) (first and fourth classes are unseen) CTMixer, MSSGU, and HZSCM; (d)–(f) (fifth and sixth classes are unseen) CTMixer, MSSGU, and HZSCM; (g)–(i) (third and sixth classes are unseen) CTMixer, MSSGU, and HZSCMT; and (j) ground-truth.

benefits from leveraging semantic information, leading to more accurate classification results.

V. Conclusion

This article presents a novel GZSL framework for HSI classification that integrates VLMs with spatial-spectral information to address zero-shot classification challenges. The framework generates pseudo-labels from VLMs using superpixel segmentation, enhances these pseudo-labels through SAM and multimodel ensembles, and refines them further by leveraging spectral information. Finally, a vanilla segmentation-based model is trained using both supervised training labels and refined pseudo-labels for ZSL in HSI classification. The proposed framework was evaluated on widely used HSI datasets, leading to the following key conclusions.

1) The proposed HZSCM framework successfully achieves zero-shot classification for HSI, demonstrating strong generalization to unseen categories. Experimental results on multiple benchmark datasets confirm its superior performance in both known and unknown classes, validating the overall effectiveness of the approach.
2) The unsupervised pseudo-label generation module effectively bridges the task paradigm gap between VLM-based region-level semantic inference and pixel-level HSI classification. The generated pseudo-labels can guide the zero-shot classification. And the ablation studies demonstrate the effectiveness of SAM and multimodel enhancement.
3) The spectral-guided pseudo-label correction module contributes to further help close the knowledge gap between VLMs and hyperspectral domain-specific features. Ablation studies demonstrate its ability to improve the quality of pseudo-labels.

This study highlights the potential of leveraging VLMs for HSI processing, suggesting a promising direction for integrating pretrained foundational models with HSI classification, particularly in open-set scenarios. Future work will focus on improving the transformation between pixel-level HSI data and region-level VLM inputs. In addition, prior knowledge from the spectral domain will be further explored to enhance collaboration with VLMs.

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